AIG: Towards a Mechanistic Science of Retrieval Integrity in Associative Memory Systems

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Abstract

Associative memory can degrade silently: internal memory geometry becomes compromised well before any retrieval output goes wrong. The Retrieval Isolation Score (RIS)—a scalar computable from memory geometry alone—drops below a warning threshold an average of 3.15±2.78 steps before behavioral failure in **94.4%** of 270 controlled trials (95% CI: 91–97%), providing a measurable window to intervene (Figure 1).

This phenomenon is particularly consequential in the MeMo project [8], which builds language model layers from explicit Correlation Matrix Memories (CMMs) [5]. Because each stored association is a separable outer product, the interference that a new write causes on existing retrievals is algebraically computable *before* the write executes—closing the certificate gap—through the Associative Inference Governor (AIG). When applied to a MeMo-consistent CMM layer, pre-execution gating and exact algebraic rollback recover retrieval quality from 0.235 to 0.957 (+0.722); under a stricter quality threshold (< 0.50), RIS warns 15.9±6.7 steps ahead in **100%** of mid-range trials; a RIS-guided retrieval blend outperforms MeMo's fixed α=0.5 blend by +109.2% in retrieval quality.

Open frontier. All results hold within single-layer CMM scope. Multi-layer certificate composition (F5)—where interference certificates do not propagate faithfully across stacked CMM layers (Section 5)—is the central open problem, not the current claim.

Keywords

associative memory governance, retrieval integrity, mechanistic interpretability, correlation matrix memory, interference certificate

1. Introduction

We observed an unexpected phenomenon in explicit associative memory: *memory systems become structurally unstable before they become behaviorally wrong.* Figure 1 illustrates the three-stage collapse in a controlled Correlation Matrix Memory (CMM): the Retrieval Isolation Score (RIS) drops first, as interfering patterns erode geometric dominance; retrieval quality drifts next as the signal diffuses; output accuracy fails last—with a predictable, measurable gap between the structural warning and the behavioral collapse. In the governed system (right panel), AIG detects the RIS warning early and blocks interfering writes, preserving accuracy throughout.

This structural precursor effect is not visible in parametric memory. Parametric systems entangle knowledge into distributed weights, making the consequences of any write algebraically opaque—perturbation of existing retrievals can only be discovered post-hoc, after generation has occurred. In explicit associative memory, failure is a continuous internal process with a measurable geometric precursor, visible inside the memory before any retrieval returns a degraded result. Unlike continual learning methods [4, 6] that respond to interference after the fact, the structural signal here appears *before* behavioral degradation. That reframing—from output event to internal dynamical state—is the central intellectual contribution of this work.

The MeMo project [8] takes a precise architectural position: building language model layers from explicit CMMs makes memory mechanistically interpretable by design. Each stored association is a separable outer product—directly readable, not hidden in distributed weights—and retrieval geometry is observable at every step. **Explicit storage is necessary but not**

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sufficient—once memory is explicit, its operations can be made formally accountable before they execute. The *certificate gap*—the blind interval between write and consequence discovery—is algebraically closable in CMMs: retrieval decomposes linearly over stored patterns, so any write's interference is computable in $O(P)$ before execution. We develop this into the Associative Inference Governor (AIG).

Contributions (CMM scope). (1) **Theory**: three governance properties of CMM structure—pre-execution certificates (P1), exact rollback (P2), empirical capacity scaling (P3, R^2=0.97). (2) **Taxonomy**: F1–F6 failure modes; four closed by AIG (F1, F2, F4, F6); F3 by key encoding; F5 open. (3) **Lead-time**: RIS precedes behavioral failure in **94.4% of 270 trials** (mean 3.15±2.78 steps); foreknowledge window confirmed across 3 sentence-embedding model families. (4) **Governance**: 0.813 vs. 0.552 accuracy (F1/F2); 97% attack-block (F6); exact rollback over 10^4 cycles (P2). (5) **MeMo**: P1+P2+adaptive α recovers quality 0.235→0.957 (+0.722); outperforms fixed α by +109.2% in retrieval quality.

2. Theoretical Framework

2.1. CMM Retrieval Geometry

Given P key-value pairs $\{(s_i, y_i)\}_{i=1}^{P}$ with normalized keys $s_i \in \mathbb{R}^d$ and values $y_i \in \mathbb{R}^m$, the Correlation Matrix Memory is $\mathbf{C} = \sum_{i=1}^{P} s_i y_i^\top$. Retrieval with query $\hat{s}$ decomposes linearly: $\hat{y} = \hat{s}^\top \mathbf{C} = \sum_{i=1}^{P} (\hat{s}^\top s_i)\, y_i$. The Retrieval Salience Value $\mathrm{RSV}_i(\hat{s}) = (\hat{s}^\top s_i)\|y_i\|$ measures each stored pattern's contribution to the retrieved output. The Retrieval Isolation Score measures whether one stored pattern dominates retrieval geometry or whether many competing patterns split the mass:

$$\mathrm{RIS}(\hat{s}) = \frac{\max_i |\mathrm{RSV}_i(\hat{s})|}{\sum_i |\mathrm{RSV}_i(\hat{s})|} \in [0, 1] \tag{1}$$

RIS → 1 indicates isolated (healthy) retrieval; low RIS indicates harmful superposition—the geometric signature of accumulating interference. Like a fault-current sensor that detects an overload before a circuit breaker trips, RIS registers structural stress before behavioral collapse.

2.2. Three Governance Properties of CMM Structure

CMM's outer-product separability makes three governance-relevant properties mathematically derivable—properties unavailable when memory is distributed in parametric weights.

P1 — Pre-execution Certificate. Storing (s_n, y_n) perturbs any existing retrieval $\hat{s}_j$ by exactly $\varepsilon_j = |\cos(s_j, s_n)| \cdot \|y_n\|$—a direct consequence of linear outer-product decomposition, computable in $O(P)$ *before* execution, tight at $\hat{s}_j = \pm s_n$. Where Ramsauer et al. [7] measure retrieval isolation post-retrieval, P1 closes the certificate gap pre-execution. Over 3100 verification trials (P=50, d=128), $\|\Delta\hat{y}_j\|$ matches ε_j to within 2×10^{-6} (float32 precision floor)—the bound is an algebraic equality, not an approximation.

P2 — Exact Rollback. Because associations are separable rank-1 outer products, $(\mathbf{C} + s_n y_n^\top) - s_n y_n^\top = \mathbf{C}$ exactly—a structural property unavailable in nonlinear parametric models [3]. Removal achieves zero perturbation and zero drift in $O(d^2)$; numerical drift is $<10^{-7}$ (float32) and exactly zero (float64) over 10^4 cycles. Under single-pattern injection at cosine 0.85: $P(\text{error}|\text{RIS}<\theta)$=1.0, $P(\text{error}|\text{RIS}\geq\theta)$=0.0, $\rho(\text{RIS}, \text{error})$=−1.0; P2 rollback at step 20 fully restores RIS.

P3 — Empirical Scaling. For random unit-sphere keys: $\mathbb{E}[\max_{j\neq n} \varepsilon_j] \approx 0.897\sqrt{2\ln P/d}$, validated at R^2=0.97 ($P\in[4, 1024]$). Operational capacity $P_{\max} = d \cdot e^{(\varepsilon^*)^2/2}$ converges to $P \approx d$ as $\varepsilon^* \to 0$, consistent with Cabannes et al. [1]. P3 is a random-geometry characterization; semantic embeddings require per-embedding recalibration—measured inflation reaches 7–9× for BERT-class models (d=768) due to cluster structure in embedding space (Table 1; §3).

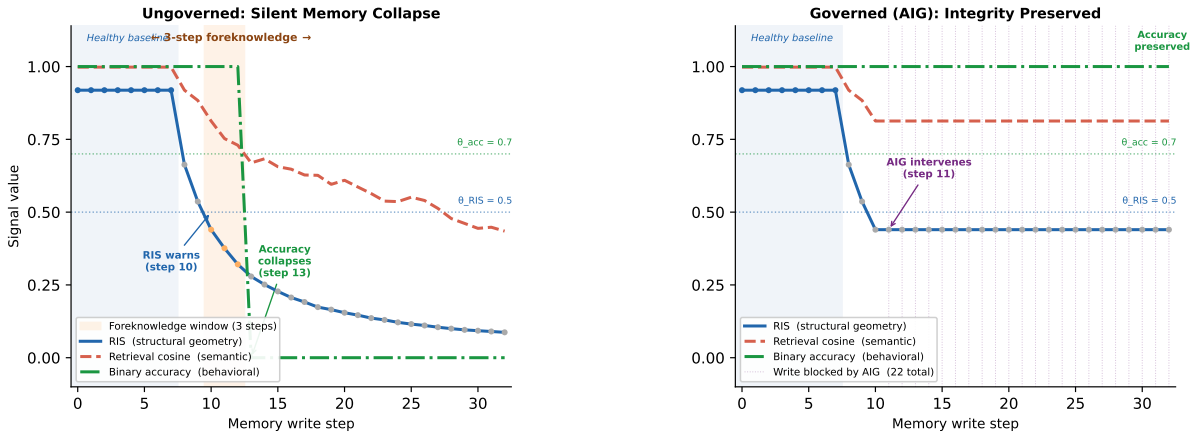


Figure 1: Hidden Memory Collapse: structural failure precedes behavioral failure—RIS detects imminent degradation before retrieval quality or accuracy deteriorates. (d=128, interferer $\cos \approx 0.42$, 25 writes attempted.) *Left:* 3-step *foreknowledge window* between RIS warning (step 10) and accuracy cliff (step 13)—memory geometrically failing while outputs appear correct. *Right:* AIG blocks 22 of 25 writes; accuracy preserved at 1.0 throughout. Ungov. final: RIS = 0.087, acc = 0.0. Gov. final: RIS = 0.440, acc = 1.0.

2.3. Governance Operations

Before each write, AIG computes ε_j for all existing patterns ($O(P)$) and estimates RIS. Three outcomes: *blocked* if $\max_j \varepsilon_j > \theta_a$ (certificate threshold) or per-anchor budget $B_j > B_{\max}=1/\theta_{\text{RIS}}-1$ (total ε admitted against anchor j so far); trust *reduced* via $\alpha(t) = \sigma(k(\text{RIS}(t)-\theta))$ if RIS $< \theta_{\text{RIS}}$; *admitted* otherwise, with P2 rollback by agent provenance (per-pattern write log). Six failure modes arise from additive outer-product superposition: F1/F2 (saturation, semantic crowding) closed by the write-gate; F3 (permutation aliasing, silent to RIS) requires circular-convolution keys; F4 (query drift) addressed by RIS-adaptive trust; F6 (gradual contamination) closed by derived budget $B_{\max}$; F5 (cross-layer amplification) is the central open problem (§5).

3. Experimental Evidence

Stability lead-time (E8). Across $d \in \{64, 128, 256\}$, interferer cosine $\approx \{0.25, 0.42, 0.58\}$, N=30 trials per condition (270 total, τ=0.75): RIS precedes behavioral failure in **94.4%** of trials (255/270; 95% CI: 91–97%), mean lead 3.15±2.78 steps—from ∼6.7 steps at low cosine (100%) to ∼0.84 at high cosine (≈83%, where interferer keys approach the anchor's geometry). The decay $\Delta \propto \mu_{\cos}^{-2.16}$ (R^2=0.979) makes the window predictable; the 3.15-step lead (τ=0.75, all regimes) and the 15.9-step lead (quality<0.50, mid-range, §4) measure the same precursor under different operational thresholds. Operationally, 3 steps is sufficient: P2 rollback or α reduction executes in $O(d^2)$, within a single memory-update cycle. *Threshold robustness:* positive rate is identical at $\theta_{\text{RIS}} \in \{0.65, 0.75\}$ (94.4% at both values); at θ=0.55 it falls to 59.3%, confirming the result is stable across the upper-threshold regime and not a single-choice artifact.

Semantic validation across model families. To confirm the foreknowledge window is not a random-geometry artifact, we test 3 sentence-embedding model families across 4 domains (60 contamination writes per trial, 6 anchor rotations per domain; θ_{RIS}=0.50, below E8's operational θ=0.65; Table 1). Across all 49 evaluated trials, the foreknowledge window is **100% positive**—structural warning precedes behavioral failure in every case. Mean Δ=26.5±2.7 steps at $\mu_{\cos} \approx 0.26$–0.28 (low semantic cosine regime): consistent with the decay law, which predicts

larger windows at lower cosine concentration. These results suggest the effect is a property of explicit associative memory architecture rather than an artifact of controlled random geometry.

Table 1
Foreknowledge window across sentence-embedding families. 3 models × 4 domains, 60 contamination writes/trial; θ_{RIS}=0.50, θ_{acc}=0.70.

Model	d	$\mu_{\cos}$	Δ (steps)	Pos. %
all-MiniLM-L6-v2	384	0.264	27.2±6.7	100%
all-mpnet-base-v2	768	0.280	29.3±10.8	100%
paraphrase-MiniLM-L6	384	0.270	22.9±5.3	100%
Overall		0.271	**26.5±2.7**	**100%**

Governance and failure mode validation. F1/F2 write-gate: 0.552→0.813 accuracy. F3 (permutation aliasing, silent to RIS): circular-convolution keys recover 50%→100%, a known scope boundary. F4: RIS-quality correlation ρ=0.938±0.019. F6 budget $B_{\max}$: 97% adversarial block rate ($\cos$ =0.45, N=60; quality 0.31→0.807). 40% RIS false-negative rate confined to dominant-wrong-key injection; 0/500 within-domain (95% CI: 0–0.7%; explicit FN tracking).

4. Extending MeMo: From Explicit Storage to Operational Accountability

MeMo's explicit outer-product architecture makes stored associations directly inspectable [8]; this work shows it also makes operations *accountable*—write consequences computable before execution rather than discovered after. At MeMo's production scale (d=2048, chunk length 384), P3 confirms that MeMo's chunk-length design keeps each CMM inside the RIS-stable capacity region.

Three capabilities MeMo's structure unlocks (GovernedMeMoLayer, a drop-in replacement for MeMo's `MeMoLayer`): *Exact removal* (P2): each outer-product association is removed algebraically exactly—to within 3×10^{-8} in float32—filling a gap in MeMo's existing `forget()` interface. *Pre-execution interference* (P1): any write's perturbation is computable in $O(P)$ before execution, algebraically unavailable in parametric models. *Continuous health* (RIS-α): a RIS-adaptive α replaces fixed α=0.5 with retrieval trust derived from live geometric state. The 40.7% block rate applies under adversarial injection.

Experimental validation (N=30 trials, d=128, interferer cosine $\in[0.25, 0.58]$; Figure 2). *(A) Predictive failure:* RIS drops below θ=0.65 an average of **15.9±6.7** steps before quality drops below 0.50 in **100%** of trials—failure is structurally visible before it is behaviorally observable. *(B) Intervention efficacy:* without governance, quality degrades to 0.235±0.087; P1 gating and P2 rollback recover it to **0.957±0.002** (+**0.722**). *(C) Adaptive alpha:* RIS-adaptive α outperforms MeMo's fixed α=0.5 by +109.2% at P=100 (transformer baseline quality 0.85).

5. Scope, Limitations, and PhD Research Plan

All results hold within single-layer CMM scope; P1 is exact in any geometric regime; P3 requires per-embedding recalibration for semantic keys. Four honest limitations:

F5: The central open problem and PhD thesis target. Single-layer certificates do not compose across MeMo layers (Figure 3): removing a pattern from layer ℓ leaves its derived key $f(s_1) = s_1^\top \mathbf{C}_1$ in layer ℓ+1 as a ghost from a prior CMM state—confirmed in MeMo's production `forget()` interface, where intermediate layers accumulate rather than subtract associations. Adversarial experiments show $\varepsilon_{L_2}/\varepsilon_{L_1}$ reaching 1.98× (mean 1.07×)—interference amplifies

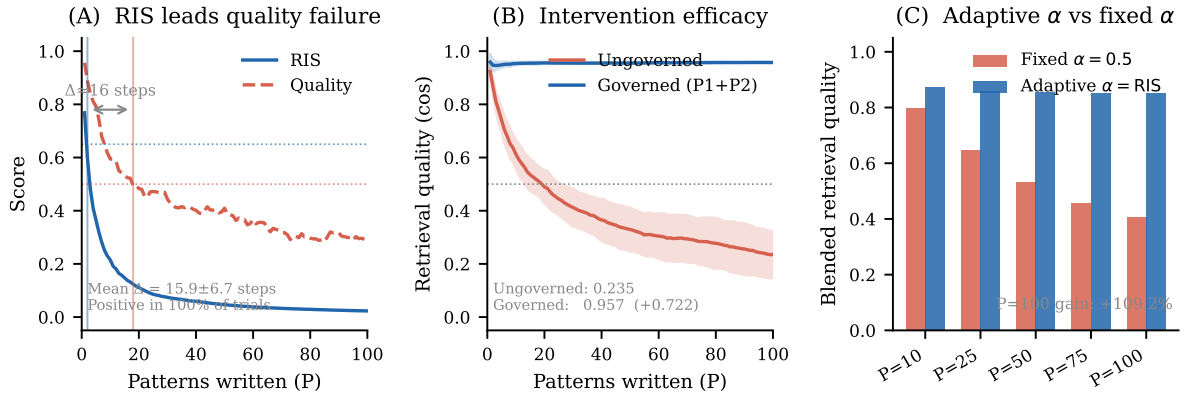


Figure 2: Operational capabilities of explicit associative memory, applied to a MeMo-style CMM layer (d=128, N=30 trials). **(A)** RIS exposes latent degradation 15.9±6.7 steps before quality failure (100% of trials). **(B)** P1+P2 provide exact interference bounds and exact rollback: quality recovers from 0.235 to 0.957 (+0.722). **(C)** RIS-adaptive α provides continuous health monitoring: +109.2% in retrieval quality over fixed α=0.5 at P=100.

Table 2
Scope and limitations.

Limitation	Description
RIS false negatives	40% under dominant-wrong-key injection; 0/500 within-domain (95% CI: 0–0.7%)
Multi-layer scope	P2 exact per layer; ghost keys persist in derived layers (F5, below)
P3 regime	Random geometry; semantic embeddings require per-embedding recalibration (7–9× measured for BERT)
CMM substrate	FFN layers are key-value memories [2]; certificate extension requires non-linear theory (Year 2+ scope)

across layers, so certificates computed for L1 underestimate downstream perturbation. F5 is not a peripheral limitation but the mathematical condition under which AIG cannot yet guarantee full accountability across MeMo's stacked layers; resolving it is the Year 1 thesis target.

PhD research plan (three phases, within the MeMo project): (1) *Year 1*—solve F5: derive a joint interference certificate for stacked CMMs and validate P2 exact rollback at MeMo's full production scale; (2) *Year 2*—test ACS (Attentional Coherence Score): the hypothesis is that a RIS-analogous concentration statistic on transformer attention weights predicts factual-recall failure before behavioral collapse on held-out QA benchmarks, via a Modern Hopfield Network bridge [7]; (3) *Year 3*—deliver a unified governance framework with formal accountability across CMMs, Hopfield networks, and attention, as deployment-ready extensions to MeMo's full language-model stack.

Contribution attribution. MeMo architecture, forgetting interface, and `directly_forget()` are due to Zanzotto et al. [8]; AIG instruments MeMo's substrate rather than modifying it. Theory, taxonomy, experiments, and F5 discovery are the PhD author's contribution, developed as an extension of MeMo's accountability theory.

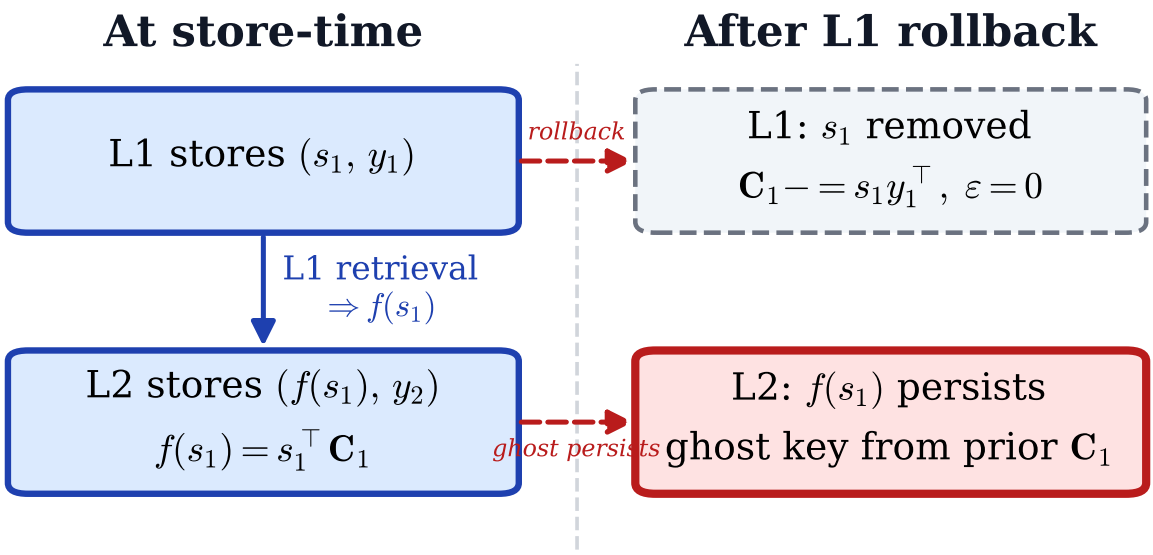


Figure 3: Multi-layer rollback limitation (F5). P2 is exact per layer. $f(s_1){=}s_1^\top \mathbf{C}_1$ is computed at store-time; after L1 rollback $\mathbf{C}_1$ is gone but $f(s_1)$ persists in L2 as a ghost key. Joint certificate composition across state-derived layers is the open problem.

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